Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins

Based on arXiv:2504.04715

For ML Researchers • AI Security Practitioners • Cloud Infrastructure Engineers

The Model Substitution Problem

Economic Incentives Drive Covert Swaps

Providers charge for access to specific models (e.g., Llama-3-70B), but the black-box nature of APIs means users have no guarantee the advertised model is actually being served.

Formal Framework:

H_0: P_actual = P_spec vs H_1: P_actual ≠ P_spec

Can an auditor distinguish the specified model's distribution from an alternative, given only API query access?

Smaller Model

e.g., 8B instead of 70B

Quantized Variant

FP8/INT8 instead of FP16

Fine-tuned Version

Updated or adapted weights

Different Family

Entirely different architecture

FP8 Quantization Preserves Most Accuracy

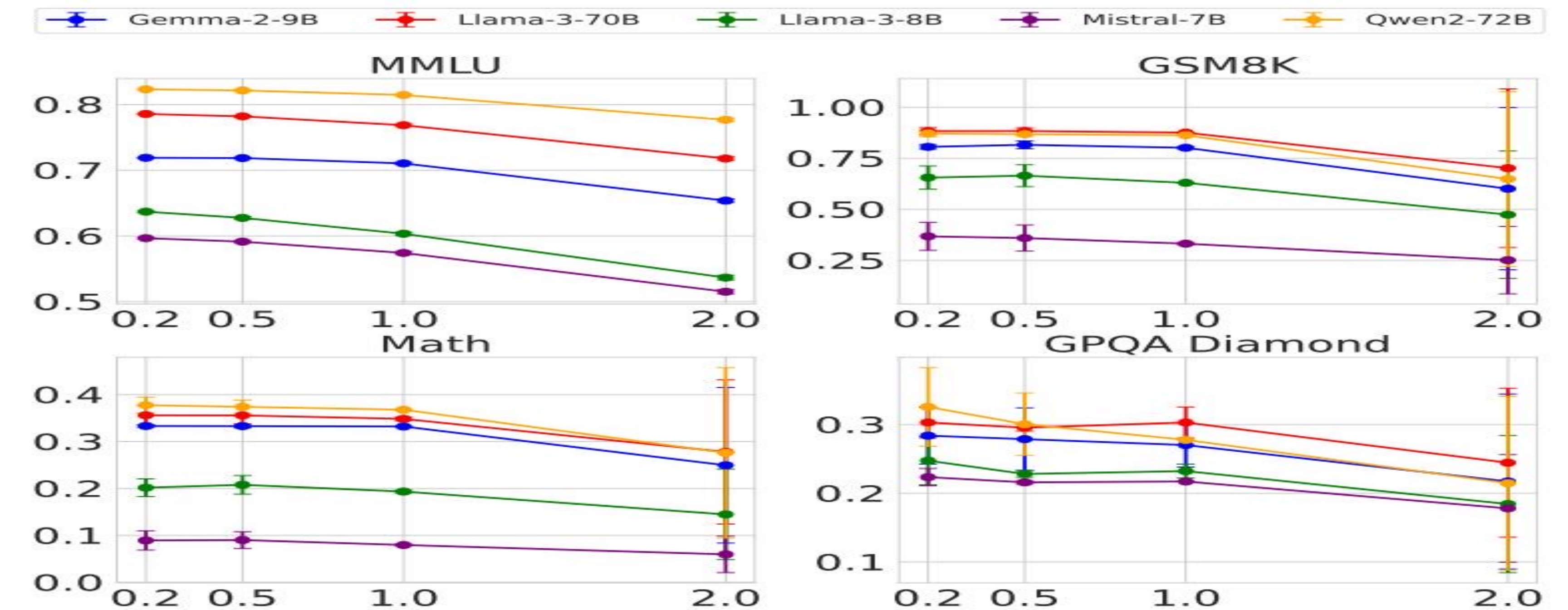
Making Substitution Hard to Detect — Performance drops are often within error bars

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
└ FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
└ FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
└ FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
└ FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
└ FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Source: arXiv:2504.04715, Table 1. Shaded rows = FP8 quantized. Most drops fall within error bars.

Higher Temperature Amplifies Gaps but Increases Noise

Benchmark accuracy vs. temperature (0.2, 0.5, 1.0, 2.0) across five model families and four tasks



At temperature 2.0, all models degrade substantially. The gap between original and quantized narrows or becomes noisy, complicating statistical detection of substitution.

Log-Probability Fingerprinting Fails

Inference Nondeterminism Defeats Metadata-Based Detection

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

The same Llama-3-8B model produces different log probabilities depending on:

- **Software framework**

transformers 4.40 vs 4.50, vLLM 0.6.2 vs 0.8.2

- **GPU hardware**

A100 vs H100 produce measurably different outputs

Auditors cannot distinguish legitimate infrastructure variation from actual model substitution.

Randomized Routing Makes Detection Query-Prohibitive

At Low Substitution Rates, Statistical Tests Need Exponentially More Queries

$$P\_mixed(y|x) = (1 - p) \cdot P\_spec(y|x) + p \cdot P\_sub(y|x)$$

As $p \to 0$, the mixed distribution converges to the specified model's distribution

p = 50%

Moderate

Detectable with standard budgets

p = 10%

~10× more

Impractical query requirements

p = 1%

~100× more

Effectively undetectable

Benchmark evasion: providers can detect known benchmark prompts and route them to the genuine model

Software-Only Auditing Methods Are Fundamentally Unreliable

Method Type	Core Weakness	Practical Limitation
Text-Based Statistical Tests (e.g., MMD)	Cannot detect subtle substitutions (FP8 scores within error bars of original)	Require impractically large query budgets; fail completely against randomized substitution at low rates
Log-Probability Fingerprinting	Production inference nondeterminism (software versions, GPU hardware) causes variation indistinguishable from substitution	Auditor cannot maintain a reliable reference; different frameworks and GPUs yield different log-probs for the same model
Adversarial Countermeasures	Providers can actively evade: benchmark evasion routes known prompts to genuine model; randomized routing mixes at low p	Detection becomes query-prohibitive as $p \to 0$; no software test can overcome an adversary who controls the serving stack

Conclusion: Software-only detection is fundamentally limited when the provider controls the inference stack. A different paradigm is needed.

TEEs Provide Provable Cryptographic Model Integrity

Trusted Execution Environments — Hardware-Secured Enclaves for Verifiable Inference

Provable

Cryptographic attestation guarantees specific model weights are loaded and correct inference code is running

Efficient

Only modest performance overhead compared to unprotected inference — practical for production deployment today

Robust

Hardware-level guarantees cannot be circumvented by the provider controlling the software stack

Why Not ZKPs?

Zero-Knowledge Proofs are computationally prohibitive for large LLMs. TEEs provide equivalent integrity guarantees with practical performance.

Deployment Path

NVIDIA Confidential Computing on H100 GPUs

Hardware-level attestation that specific model weights are loaded and executed with correct code

Available today — no fundamental research barriers remain

Key Takeaways

Close the Verification Gap with Hardware Attestation

01 Model substitution is economically motivated and practically undetectable via software alone

02 FP8 quantization reduces costs significantly while barely affecting benchmark scores — an attractive, nearly invisible substitution

03 Inference nondeterminism across software versions and GPU hardware defeats log-probability fingerprinting

04 Randomized substitution routing makes all statistical detection methods query-prohibitive

05 Trusted Execution Environments are the only currently viable path to verifiable LLM API integrity

The community must push for hardware attestation as a standard requirement for commercial LLM API providers.